



Aalto University
School of Chemical
Engineering

Phonon transport properties

MSSC2025, Torino, 2025-09-12

Antti Karttunen

Aalto University, Finland

Department of Chemistry and Materials Science

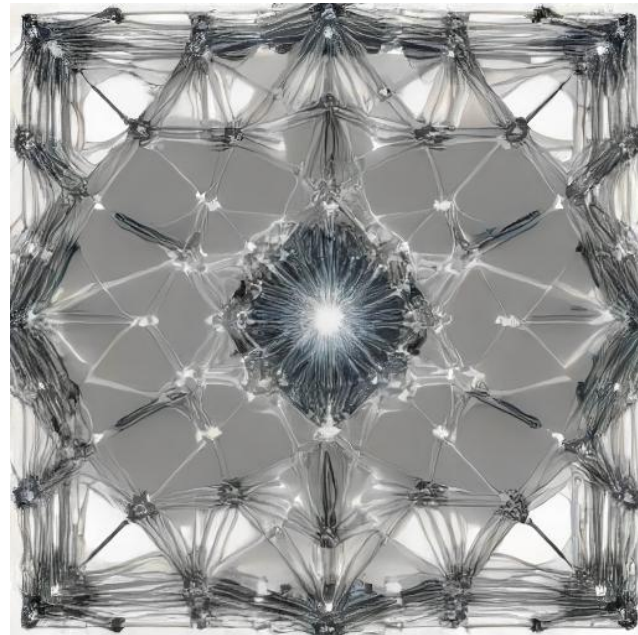
Closely related talks at MSSC2025

Wednesday lecture on *Harmonic Lattice Dynamics* by Anna Ferrari

Wednesday lecture on *Thermodynamics Beyond the Harmonic Approximation* by Alessandro Erba

Wednesday lecture on *Anharmonic Vibrational States and Spectra* by Davide Mitoli

Friday lecture on *Strongly Anharmonic Systems* by Matteo Calandra



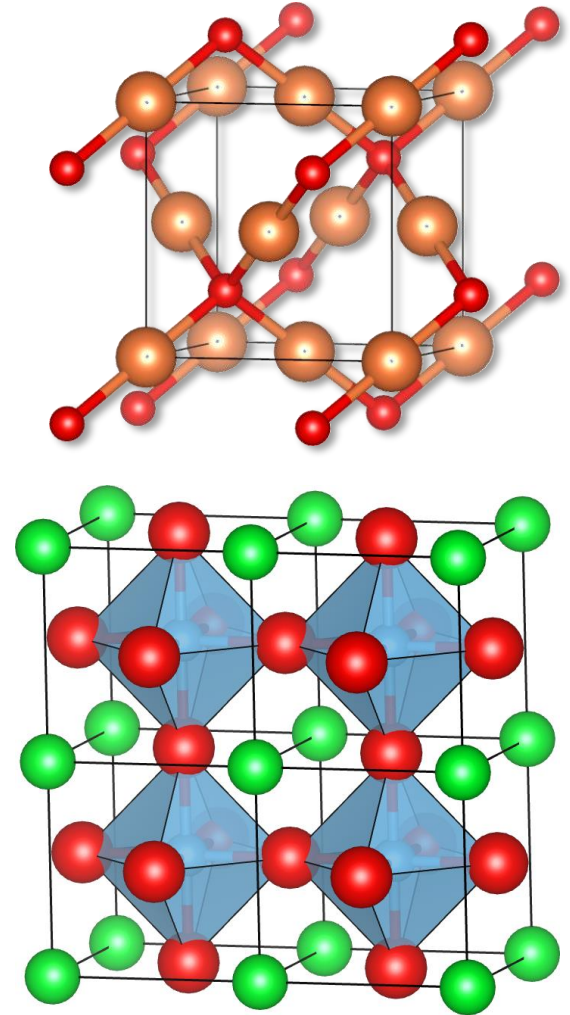
Generated with deepai.org using text prompt
"Crystal structure with anharmonic phonon modes"

Outline

Introduction to phonon transport and lattice thermal conductivity.

Lattice thermal conductivity with CRYSTAL and Phono3py.

Finite-temperature phonons with CRYSTAL and ALAMODE.



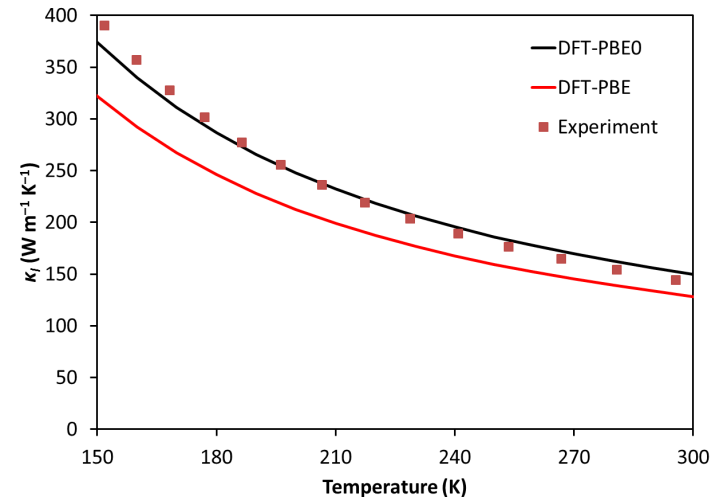
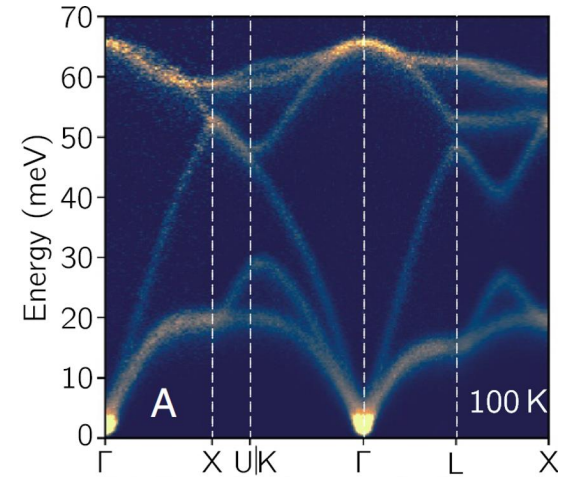
Phonons

Phonons and phonon properties

Vibrational excitations in a solid are called *phonons*

Examples of material properties that depend on phonons:

- Thermodynamics
- Vibrational spectroscopy (IR, Raman, Inelastic Neutron Scattering)
- Thermal expansion
- Thermal conductivity
- Pyroelectricity



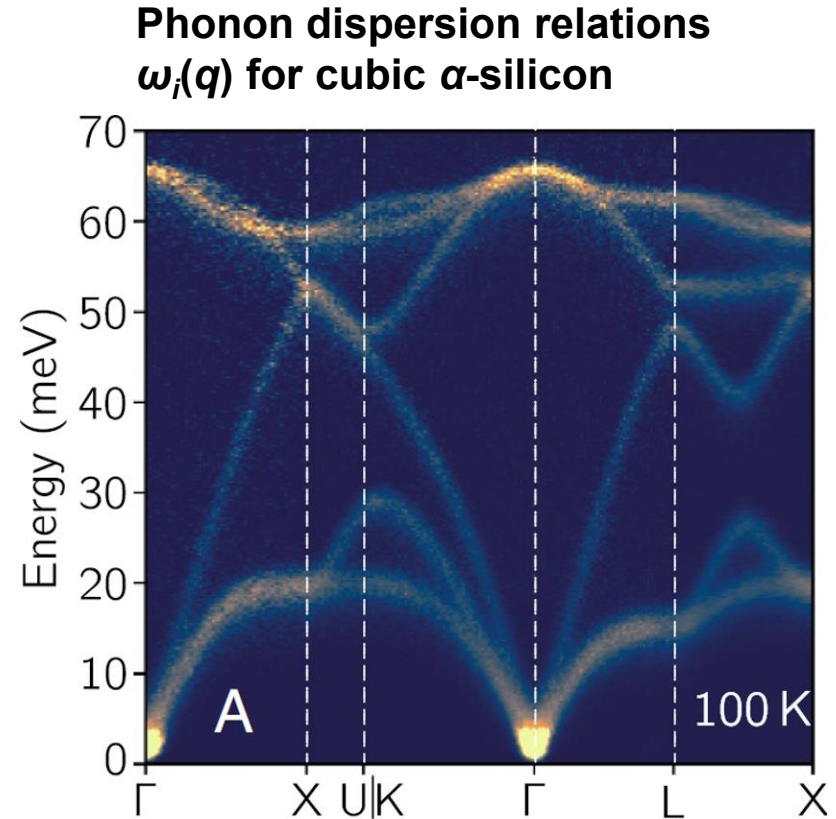
Vibrational modes in the solid state

Phonon energy ω_i is described as a function of wavevector q in the reciprocal space

$\omega_i(q)$ = *phonon dispersion relations*

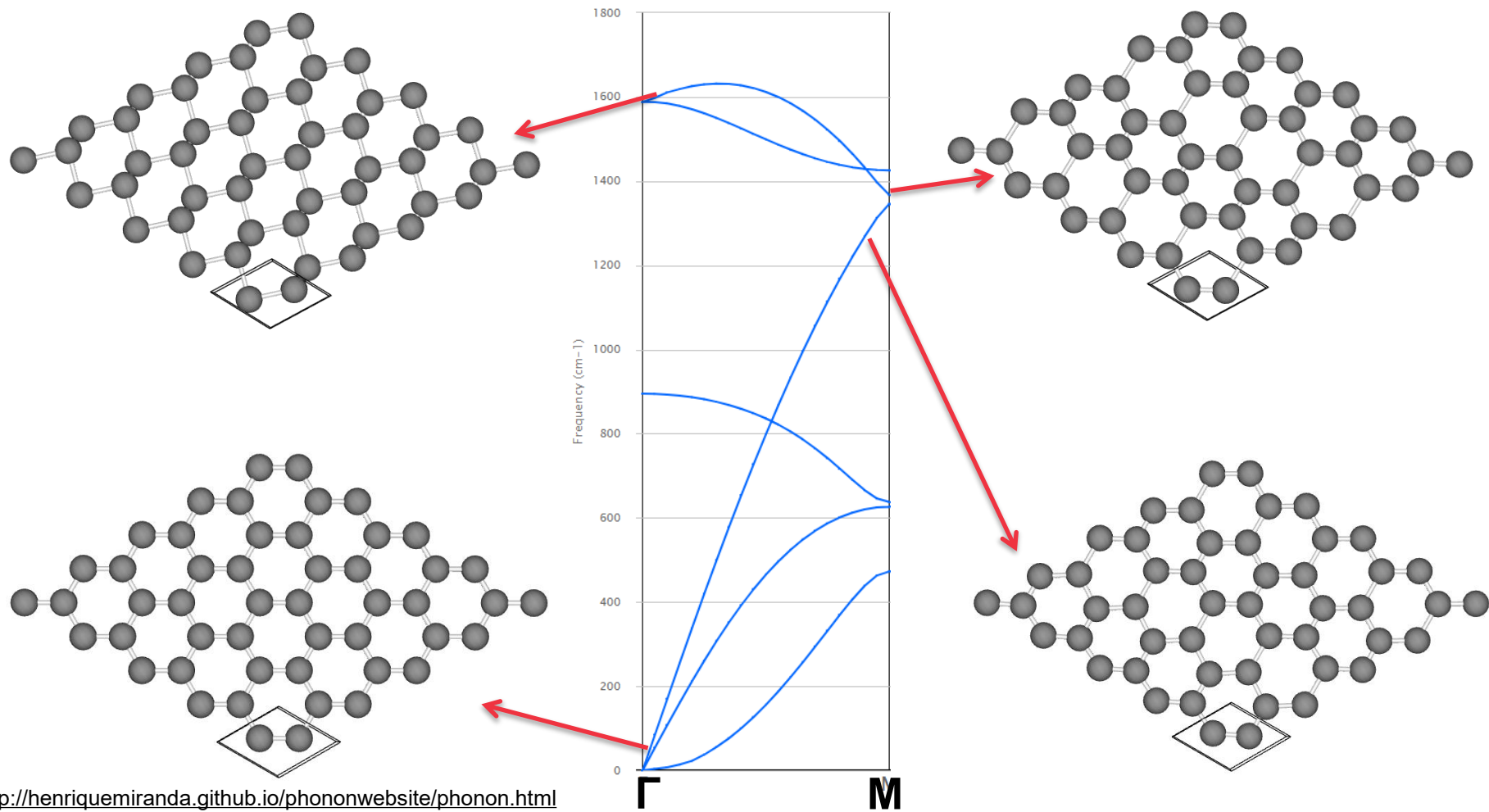
Phonon dispersion relations account for the periodic nature of the crystal

Atoms in neighboring cells can vibrate in-phase or out-of-phase



Inelastic neutron scattering data at 100 K
(*Proc. Natl. Acad. Sci.* 2018, 115, 1192)

Some phonons for graphene (2D)



Workflow for phonon dispersions (α -Si)

1. Optimize the geometry of a solid

- For α -Si, only the lattice constant a needs to be optimized

2. Create supercells with finite displacements using Phonopy

- Real-space supercells enable the sampling of the $\mathbf{q}$ -space
- Here, a $3 \times 3 \times 3$ supercell of the crystallographic cell (216 atoms)
- For α -Si, it is enough to calculate one supercell (high symmetry)

3. Calculate forces for all supercells with CRYSTAL

4. Calculate 2nd order force constants from numerical 2nd derivatives with Phonopy

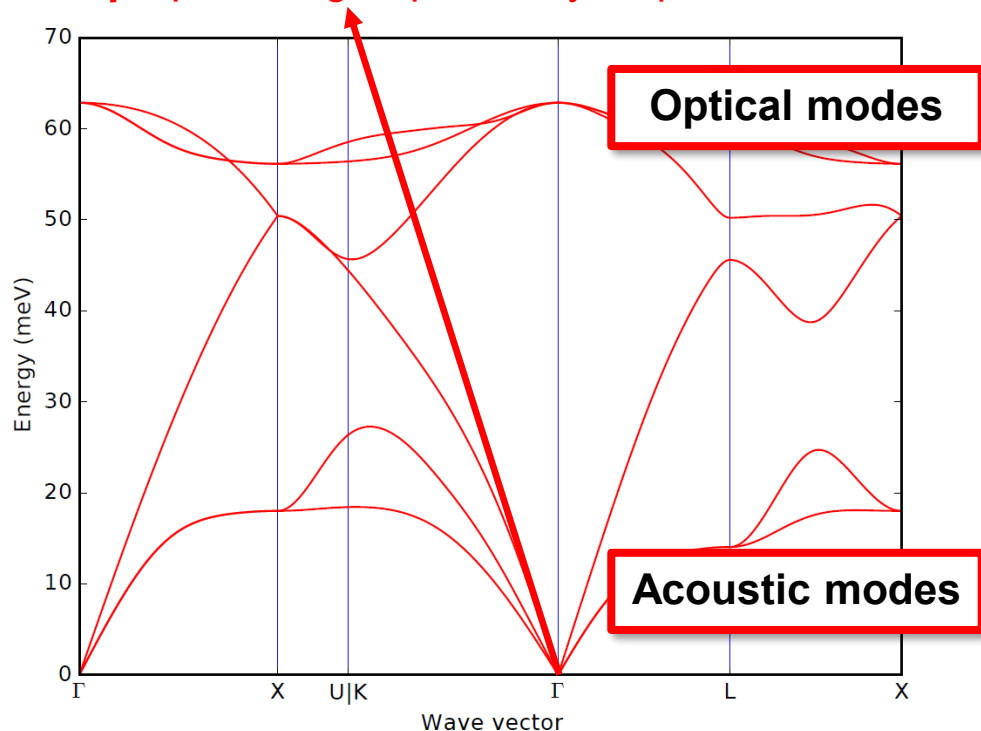
5. Calculate phonon dispersion relations with Phonopy using lattice dynamics (harmonic approximation)

¹ Atsushi Togo, Laurent Chaput, Terumasa Tadano, and Isao Tanaka, *J. Phys. Condens. Matter* 35, 353001-1-22 (2023), <https://phonopy.github.io/phonopy>

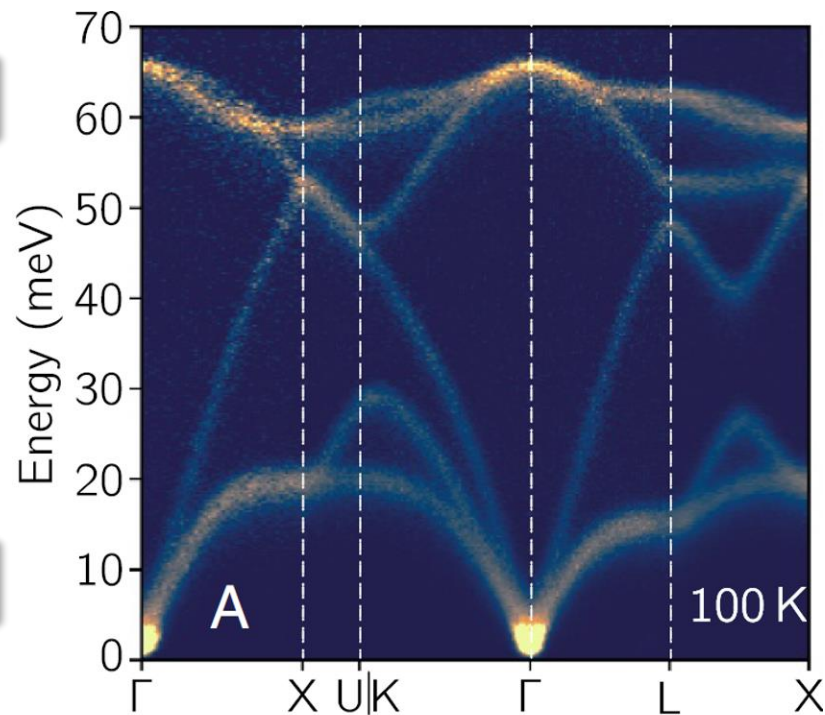
² Atsushi Togo, *J. Phys. Soc. Jpn.*, 92, 012001-1-21 (2023), <https://phonopy.github.io/phono3py/>

Comparison with experiment (α -Si)

$d\omega/dq$ = phonon group velocity $\approx$ speed of sound



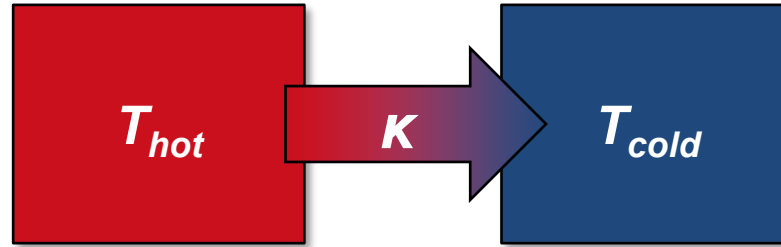
CRYSTAL + Phonopy (0 K, DFT-PBE0/SVP)



Inelastic neutron scattering data at 100 K
(*Proc. Natl. Acad. Sci.* 2018, 115, 1192)

Thermal conductivity

Thermal conductivity κ



Ability of a substance to conduct heat (units: $\text{W m}^{-1} \text{K}^{-1}$)

Very important material property in a number of fields:

- Microelectronics (thermal dissipation)
- Thermal barrier coatings in gas turbines (thermal insulation)
- Thermoelectrics (direct heat-to-electricity conversion)

Warm-up: High or low thermal conductivity?



Diamond



Graphite



Figures:
Wikipedia

Polyethylene



Silicon



Silica aerogel

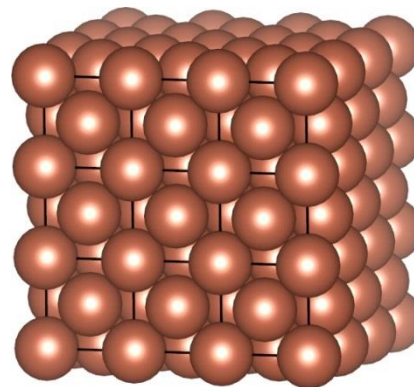


Figure: AJK

Copper

Atomic-level mechanisms behind thermal conductivity

Metals

- Heat is primarily carried by ***conduction electrons***

Insulators

- Heat is primarily carried by ***phonons*** (vibrational excitations)

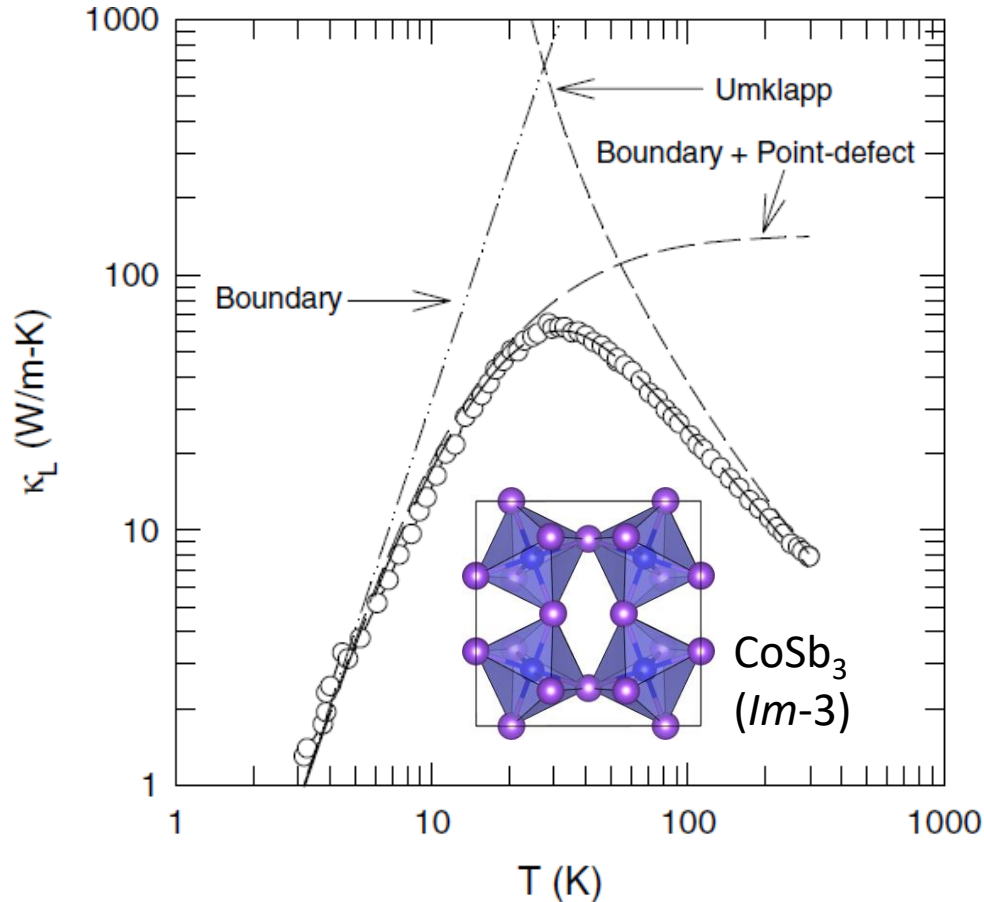
Semiconductors

- Low doping levels → insulator-like behavior
- High doping levels → both phononic and electronic contribution

Focus of the lecture:
Lattice thermal conductivity in electrical insulators

Thermal conductivity of an insulator

Highly temperature-dependent, limited by phonon scattering



Circles = experiment (CoSb_3)

Dashed curves = Theoretical limits imposed by various phonon scattering mechanisms:

- Boundary
- Point-defect / mass-defect (isotope)
- Umklapp = **phonon-phonon** scattering (anharmonicity)
- At $T > 100$ K, phonon-phonon scattering typically dominates ($1/T$ dependence)

Phonon scattering

Materials have finite thermal conductivity because of various phonon *scattering mechanisms*

- Boundary, point-defect, isotope, phonon-phonon, electron-phonon

Phonon-phonon scattering is the most important scattering mechanism for crystalline, electrically insulating materials at the room temperature

- Crystal with only harmonic phonons would have no phonon–phonon scattering
- Phonon-phonon scattering arises due to ***anharmonicity*** (deviation from harmonic oscillator behavior)

Lattice thermal conductivity with CRYSTAL and Phono3py

Lattice thermal conductivity κ , from Boltzmann Transport Equation

κ can be solved with the help of the semiclassical Boltzmann Transport Equation (BTE)

In Single-Mode Relaxation Time approximation:

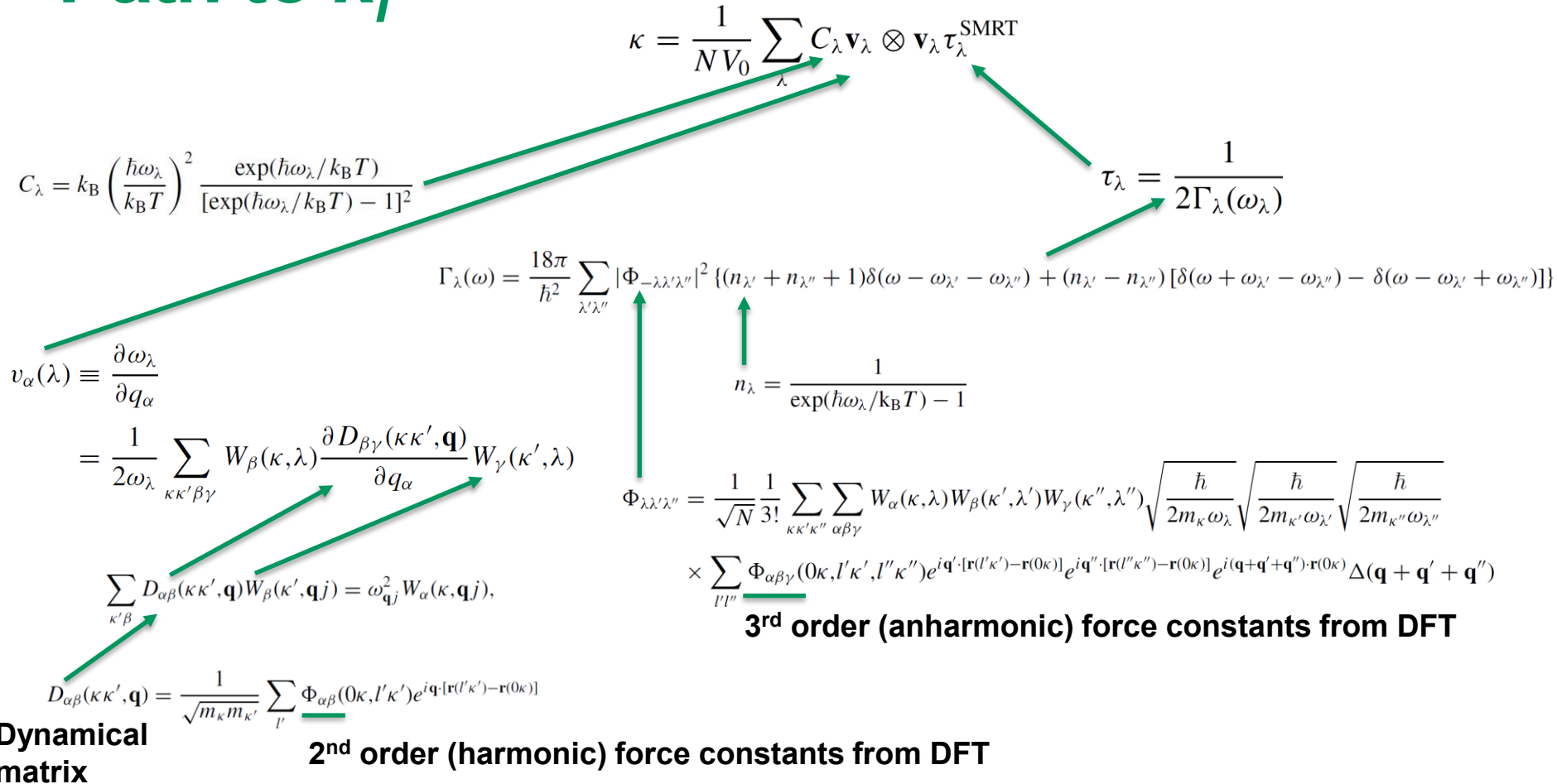
$$\kappa = \frac{1}{NV_0} \sum_{\lambda} C_{\lambda} \mathbf{v}_{\lambda} \otimes \mathbf{v}_{\lambda} \tau_{\lambda}^{\text{SMRT}}$$

Heat capacity,
calculated from
phonon
dispersions
[$\lambda = (q, j)$]

Phonon group
velocity, calculated
from phonon
dispersions

Phonon relaxation time, calculated
from phonon scattering rates (using
phonon dispersions and **anharmonic**,
third-order force constants)

Path to κ



Finite-displacement-based numerical approach to obtain κ ,

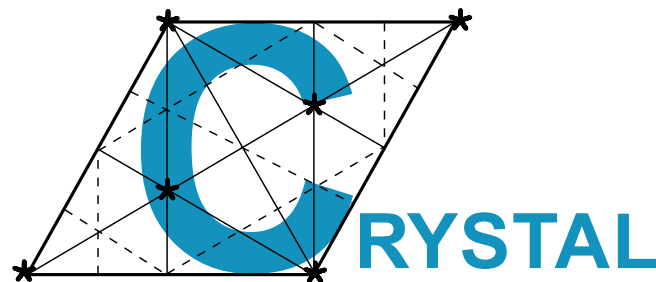
Harmonic phonon properties:
Phonopy^[1, 2]



Lattice thermal conductivity
(BTE): Phono3py^[1, 2]



DFT calculations: CRYSTAL



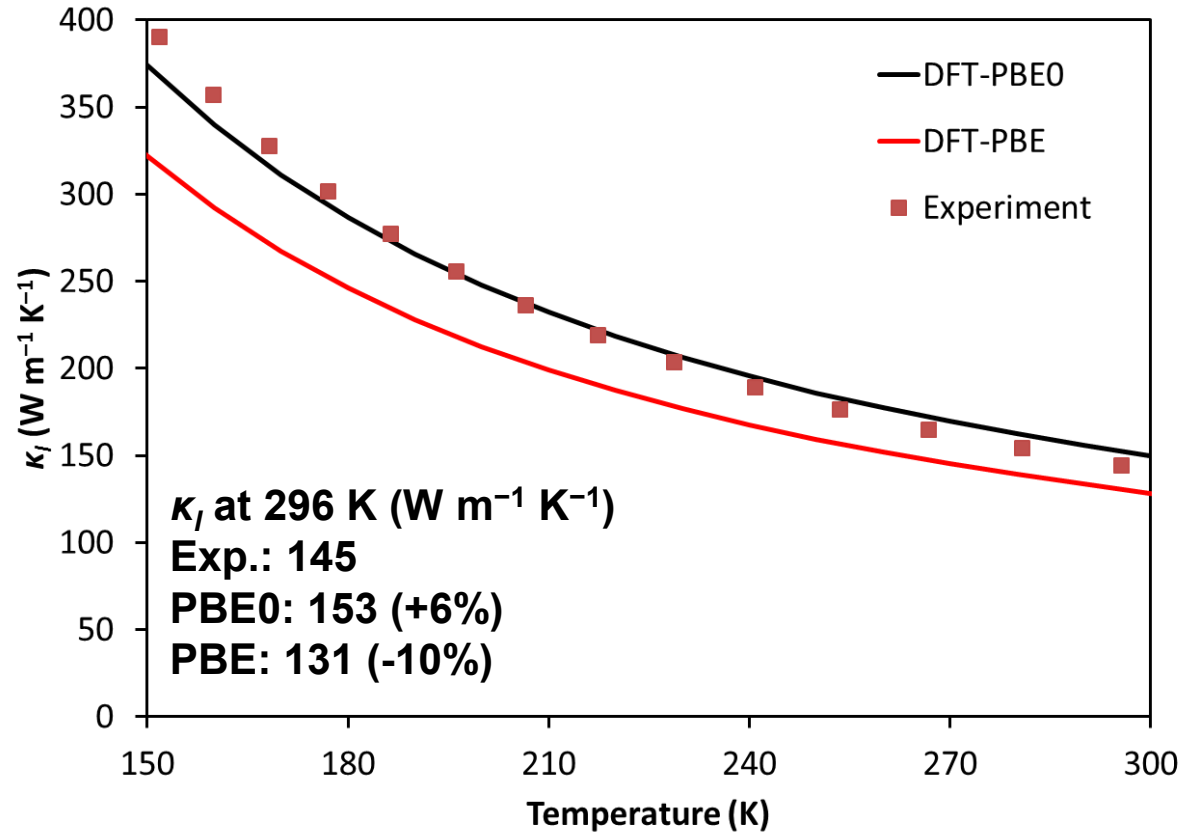
¹ Atsushi Togo, Laurent Chaput, Terumasa Tadano, and Isao Tanaka, *J. Phys. Condens. Matter* 35, 353001-1-22 (2023), <https://phonopy.github.io/phonopy> and <https://phonopy.github.io/phono3py/>

² Atsushi Togo, *J. Phys. Soc. Jpn.*, 92, 012001-1-21 (2023)

Workflow for obtaining κ_l (ZnO)

- 1. Optimize the geometry of a solid with CRYSTAL**
 - Check Γ -point frequencies before proceeding!
- 2. Create supercells with finite displacements using Phono3py**
 - Real-space supercells enable the sampling of the $\mathbf{q}$ -space
 - 2nd order force constants (6 supercells, 4x4x3, 192 atoms)
 - 3rd order force constants (1254 supercells, 3x3x2, 72 atoms)
- 3. Calculate forces (first derivatives of energy) for all supercells with CRYSTAL**
- 4. Calculate 2nd and 3rd order force constants from numerical 2nd and 3rd derivatives with Phono3py**
- 5. Calculate phonon dispersions and lattice thermal conductivity with Phono3py (BTE)**

α -Silicon lattice thermal conductivity



SVP basis set. Supercells 6x6x6 (2nd) and 3x3x3 (3rd), 30x30x30 $\mathbf{q}$ -mesh for BTE-RTA, isotope scattering
Experiment: A. V. Inyushkin et al. *Phys. Status Solidi C* **2004**, 1, 2995–2998.

Beyond defect-free single crystals

Polycrystalline materials can be modelled with a simple boundary scattering model (grain size limited phonon transport)

Effect of electron-phonon scattering and point defects in ongoing development

Journal of
Materials Chemistry A



PAPER

[View Article Online](#)

[View Journal](#) | [View Issue](#)



Cite this: *J. Mater. Chem. A*, 2020, 8, 1273

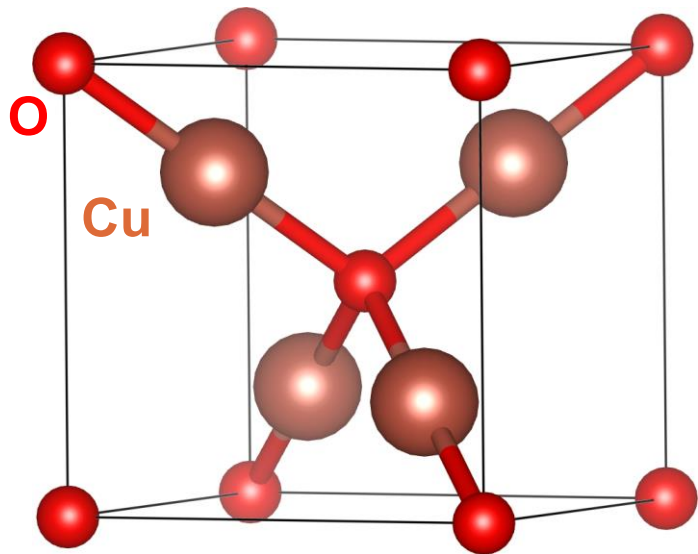
Combined treatment of phonon scattering by electrons and point defects explains the thermal conductivity reduction in highly-doped Si

Bonny Dongre, ^{*a} Jesús Carrete, ^a Shihao Wen, ^b Jinlong Ma, ^b Wu Li, ^b Natalio Mingo ^c and Georg K. H. Madsen ^a

Example: Cu_2O



Computational details for Cu₂O



Cu₂O (*Pn-3m*)
6 atoms in the unit cell

PBE0/TZVP and PBE/TZVP

Lattice parameter
(experiment: 4.269 Å at $T = 295$ K)

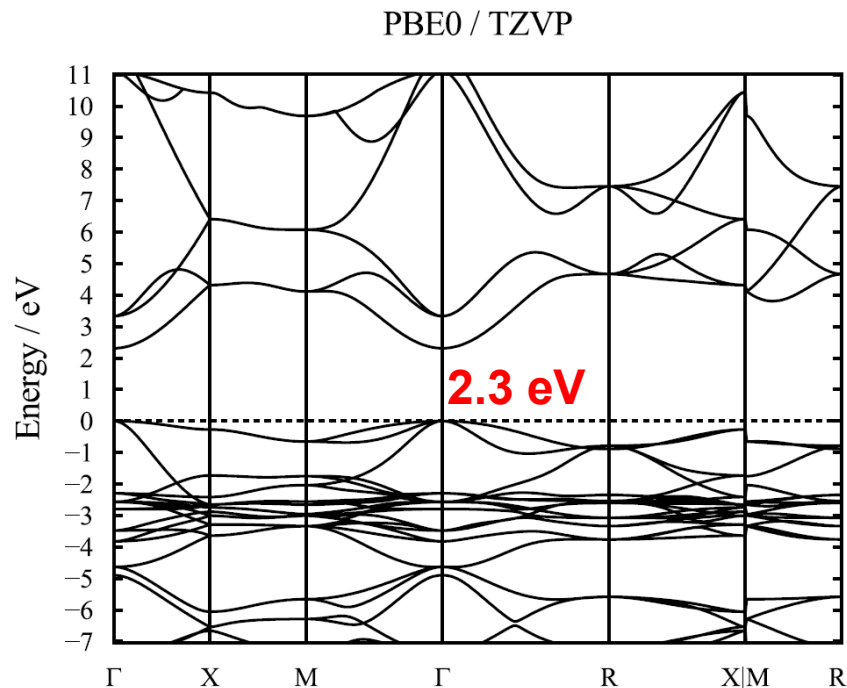
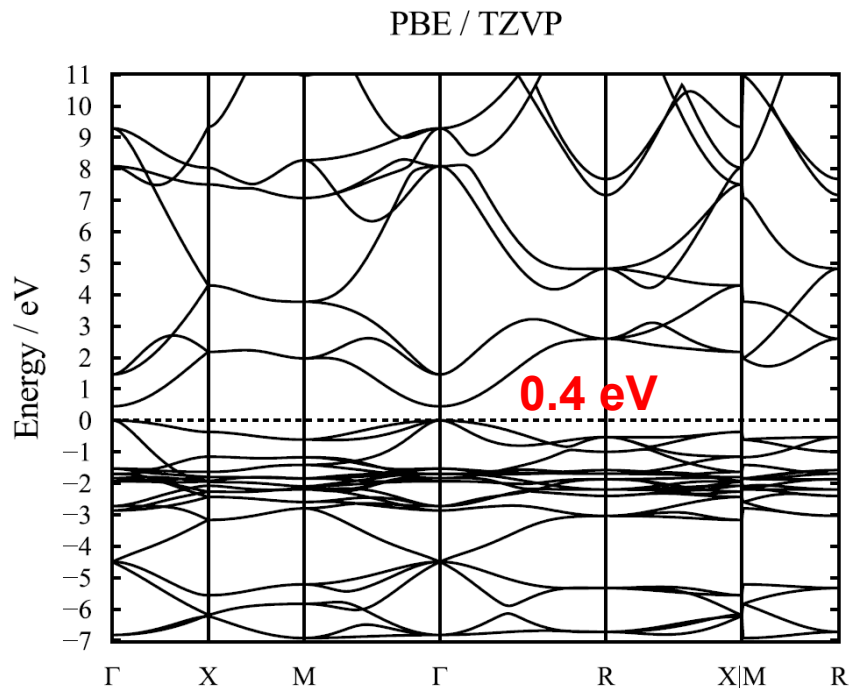
	SVP	TZVP
PBE	4.303 (+0.81%)	4.328 (+1.39%)
PBE0	4.296 (+0.64%)	4.318 (+1.16%)
HSE06	4.296 (+0.64%)	4.318 (+1.16%)

Harmonic force constants: 4x4x4
supercell with 384 atoms

Anharmonic FCs: 2x2x2 supercell
with 48 atoms, no distance cutoffs

¹ J. Linnerra, A. J. Karttunen, *Phys. Rev. B* **2017**, 96, 014304.

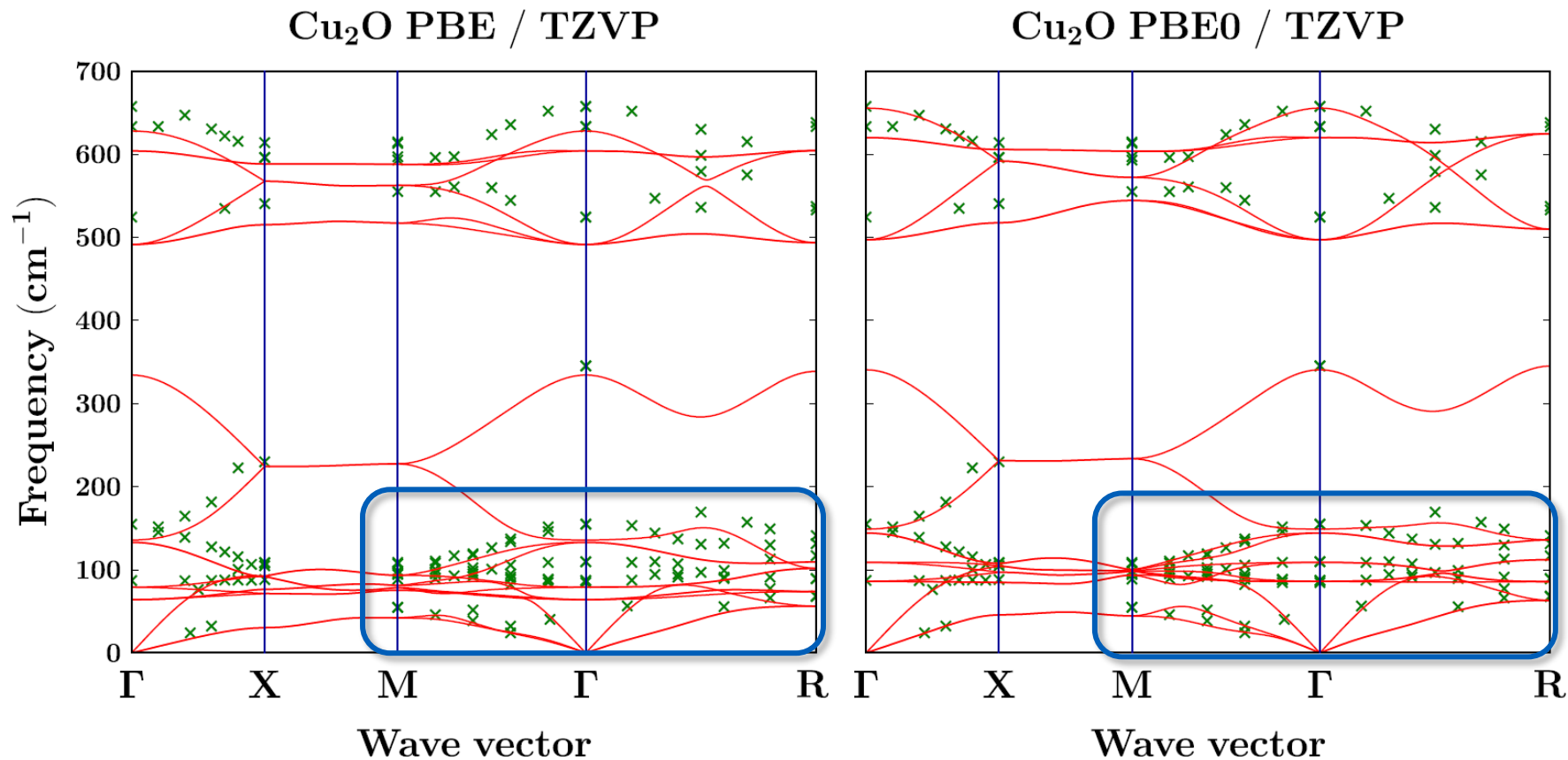
Cu₂O electronic band structures



Experimental band gap (0 K) is 2.17 eV^[1]

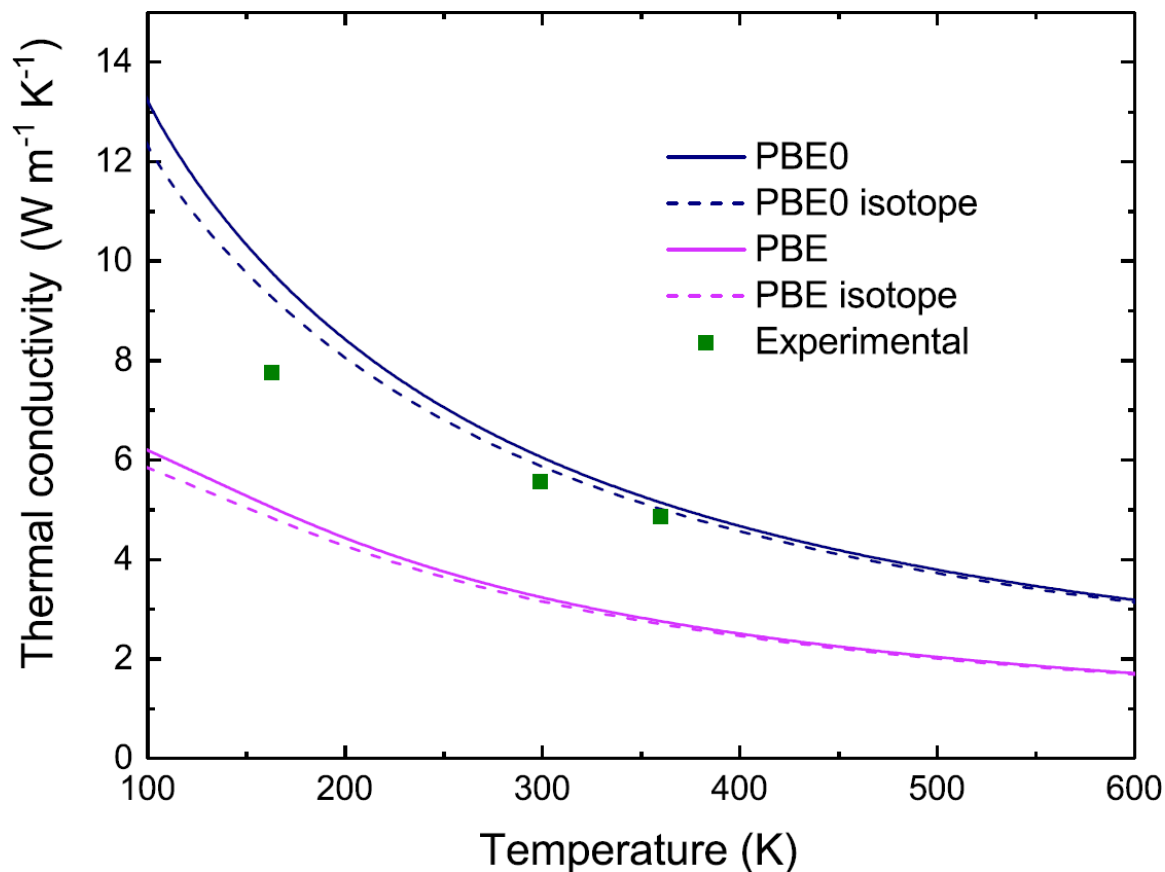
¹ C. Kittel, Introduction to Solid State Physics, 8th ed., p. 190.

Cu₂O phonon dispersions



¹ **Exp. data:** K.-P. Bohnen, R. Heid, L. Pintschovius, A. Soon, C. Stampfl, *Phys. Rev. B* **2009**, 80, 134304.

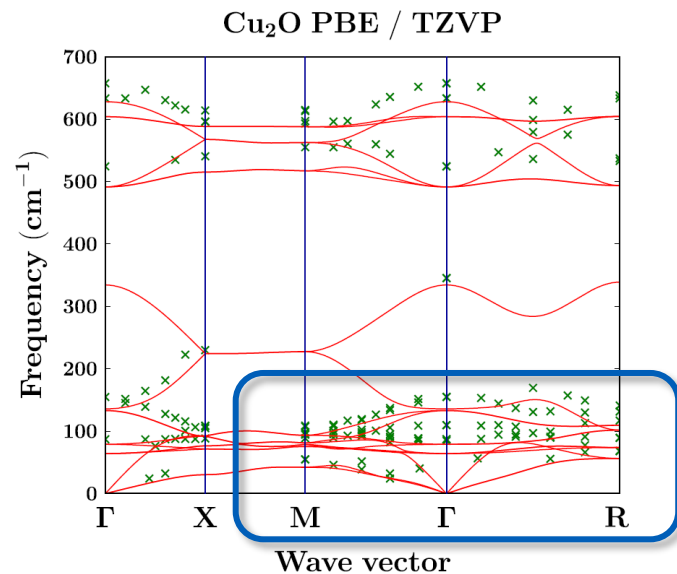
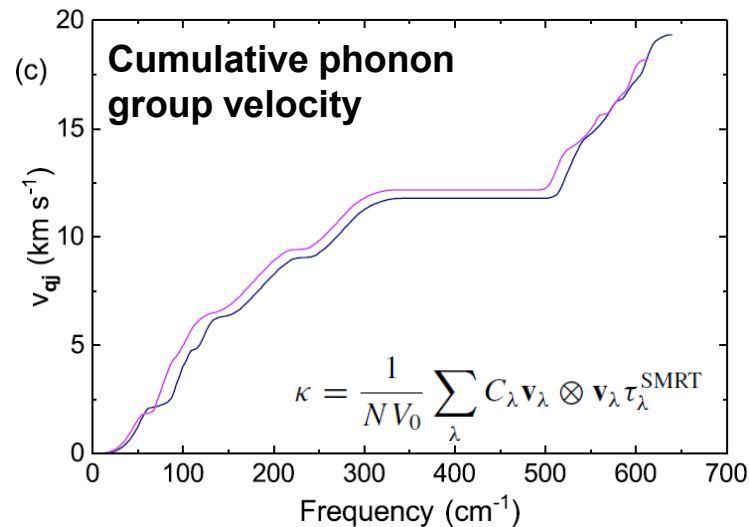
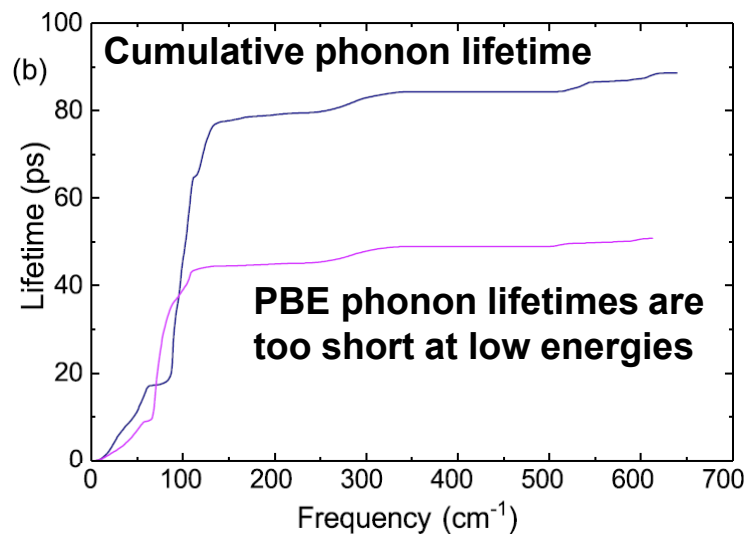
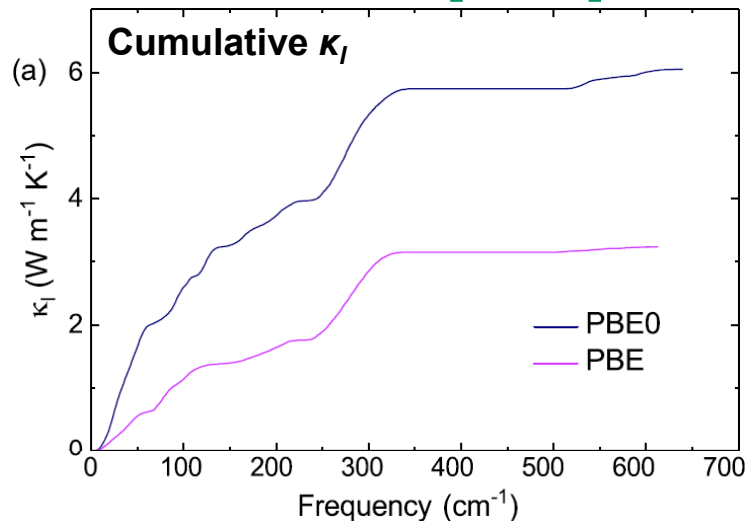
Cu₂O lattice thermal conductivity



κ_l at 300 K ($\text{W m}^{-1} \text{K}^{-1}$)
Exp.: 5.6
DFT-PBE0/TZVP: 5.9 (+5%)
DFT-PBE/TZVP: 3.2 (-44%)

¹ J. Linnerna, A. J. Karttunen, *Phys. Rev. B* **2017**, 96, 014304.

Cumulative properties



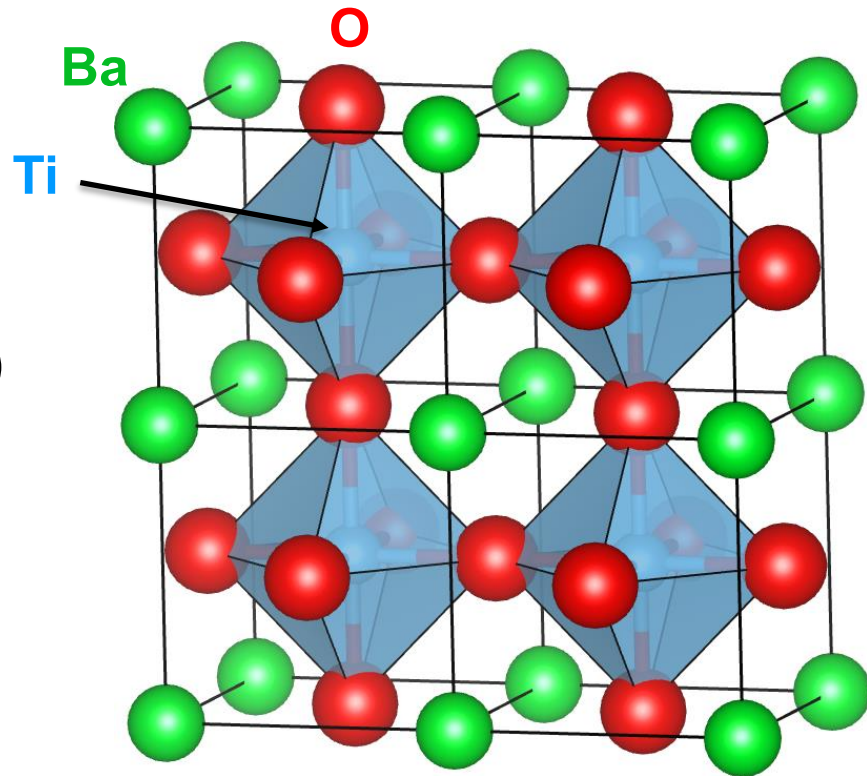
Finite-temperature phonons with CRYSTAL and ALAMODE

Benchmark system: Ferroelectric BaTiO_3 perovskite

BaTiO_3 (tetragonal at 300 K, $P4mm$).

Strongly anharmonic material.

Harmonic phonon dispersions (0 K) show imaginary phonon modes.

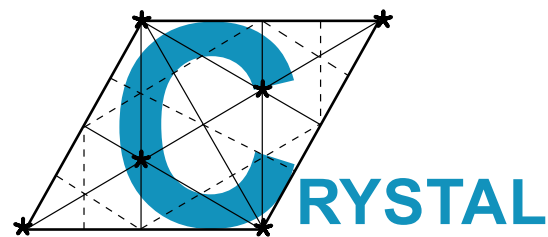


Anharmonic finite-temperature phonon properties

Anharmonic phonon properties with
ALAMODE.^[1,2]

ALAMODE needs a quantum chemical
code to calculate energies and forces.

ALAMODE + CRYSTAL interface by Kim
Eklund (Aalto)^[3, 4]



¹ T. Tadano, Y. Gohda, and S. Tsuneyuki, *J. Phys.: Condens. Matter* **26**, 225402 (2014).

² T. Tadano and S. Tsuneyuki, *Phys. Rev. B* **92**, 054301 (2015).

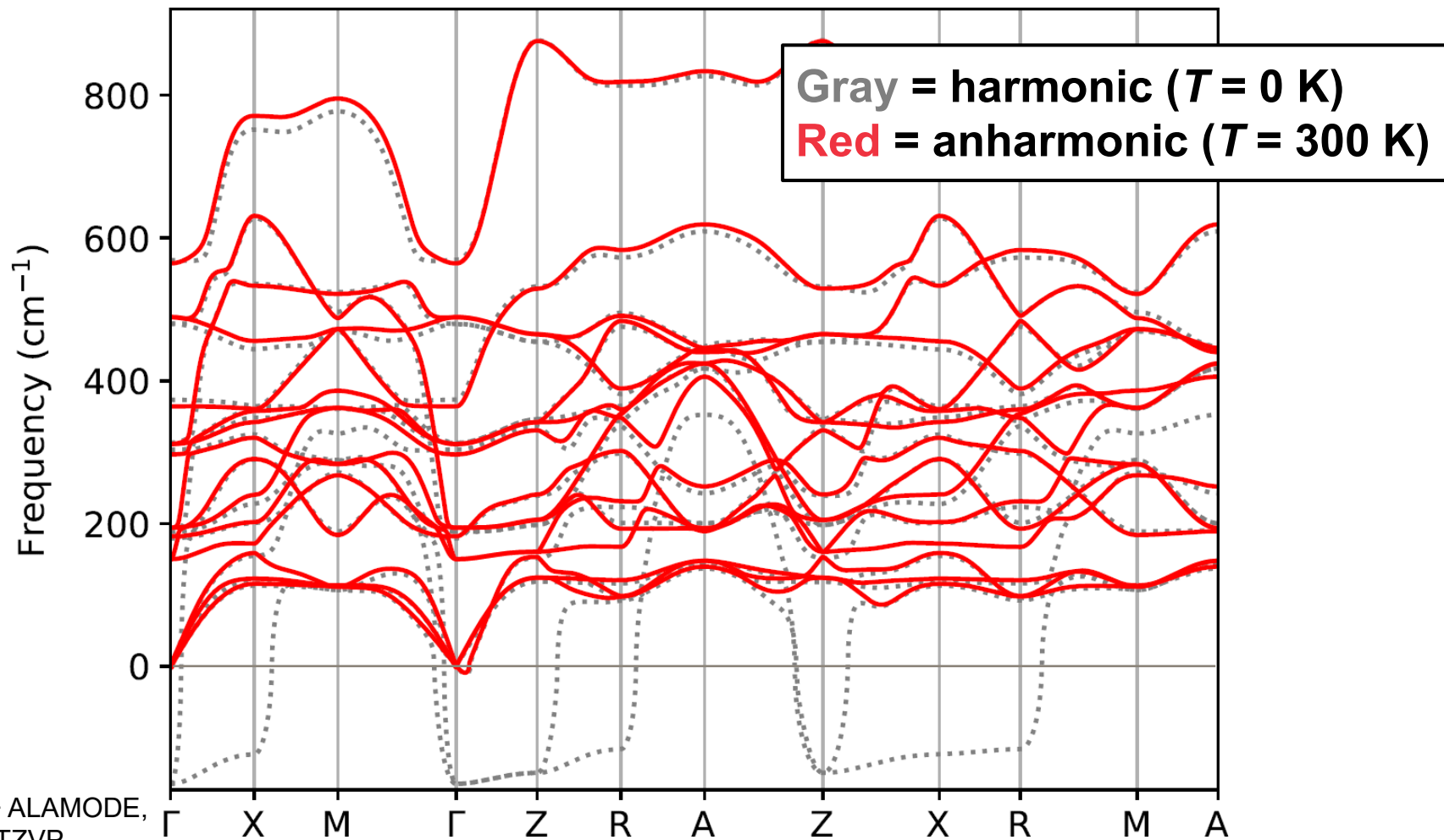
³ K. Eklund, A. J. Karttunen, *J. Phys. Chem C* **2023**, *127*, 21806–21815.

⁴ K. Eklund, A. J. Karttunen, *J. Phys Chem. C* **2024**, *128*, 16199–16207.

Workflow for finite-temperature phonons

1. **Optimize the structure of a solid with CRYSTAL.**
2. **ALAMODE creates supercells with random displacements.**
 - Real-space supercells enable the sampling of the phonon $\mathbf{q}$ -space
 - 40 supercells of $3 \times 3 \times 3$ size were enough for BaTiO_3 (135 atoms)
3. **Calculate forces (first derivatives of energy) with CRYSTAL.**
4. **ALAMODE obtains interatomic force constants with regression methods (2nd, 3rd, 4th order or even higher).**
5. **ALAMODE first calculates harmonic phonon dispersions and then the finite-temperature phonon frequencies on top of that with Self-Consistent Phonon Theory.**

BaTiO₃ phonon dispersions



Lattice thermal conductivity of ferroelectric PbTiO_3

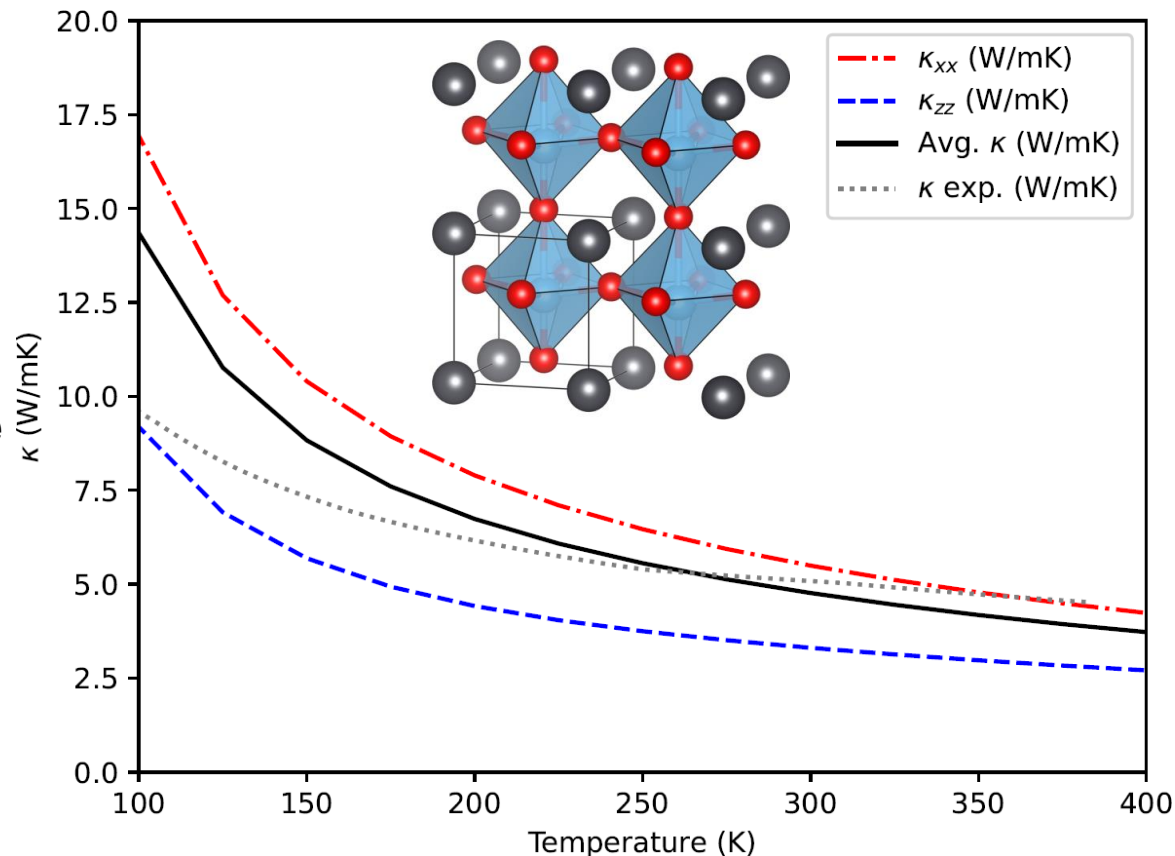
Crystal structure at 300 K is tetragonal ($P4mm$), similar to BaTiO_3

CRYSTAL + ALAMODE

DFT-PBEsol0/TZVP

BTE-RTA with 4th order force constants

Isotope scattering included



Summary

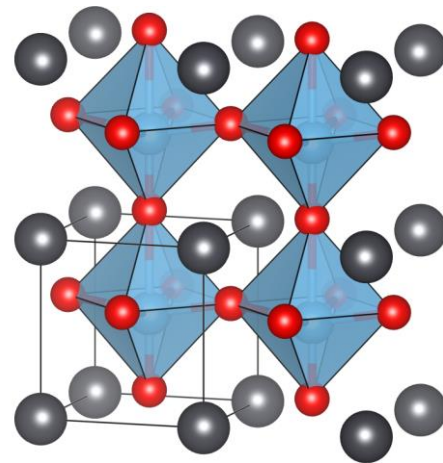
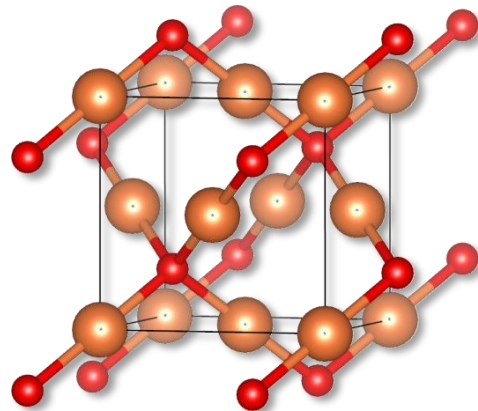
Summary

Lattice thermal conductivity of crystalline solids can be investigated with CRYSTAL+Phono3py.

The possibility to study lattice thermal conductivity with hybrid DFT methods is crucial for *d*-metal oxides.

Phonopy and Phono3py are free and open source. Documentation and CRYSTAL examples are included.

Finite-temperature phonon dispersions can be studied with CRYSTAL+ALAMODE.



Acknowledgments

Studies with ALAMODE

Kim Eklund (Aalto University)

Funding:

Pyroelectric Materials for Thermal Energy
Harvesting (RCF/363733)

Computational resources:

CSC — the Finnish IT Center for Science

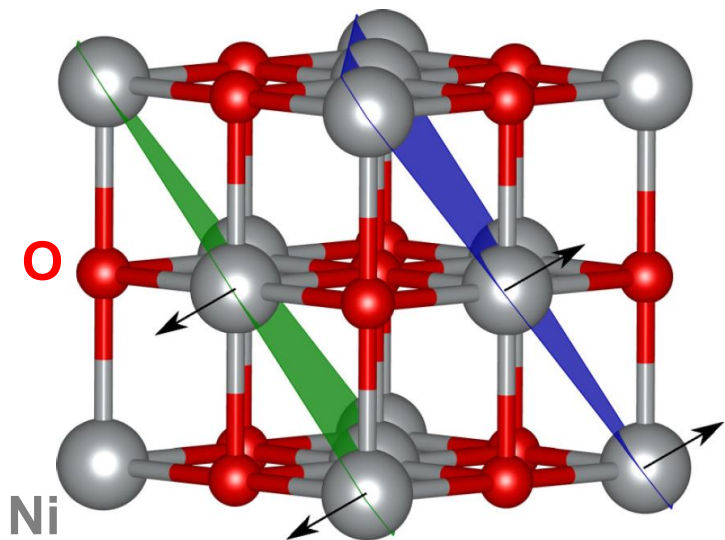


**Research Council
of Finland**



Appendix 1: NiO

Nickel(II) oxide, NiO



**Level of theory:
DFT-PBE0/TZVP**

Non-magnetic primitive cell: Ni + O ($Fm-3m$)

NiO is antiferromagnetic up to 525 K

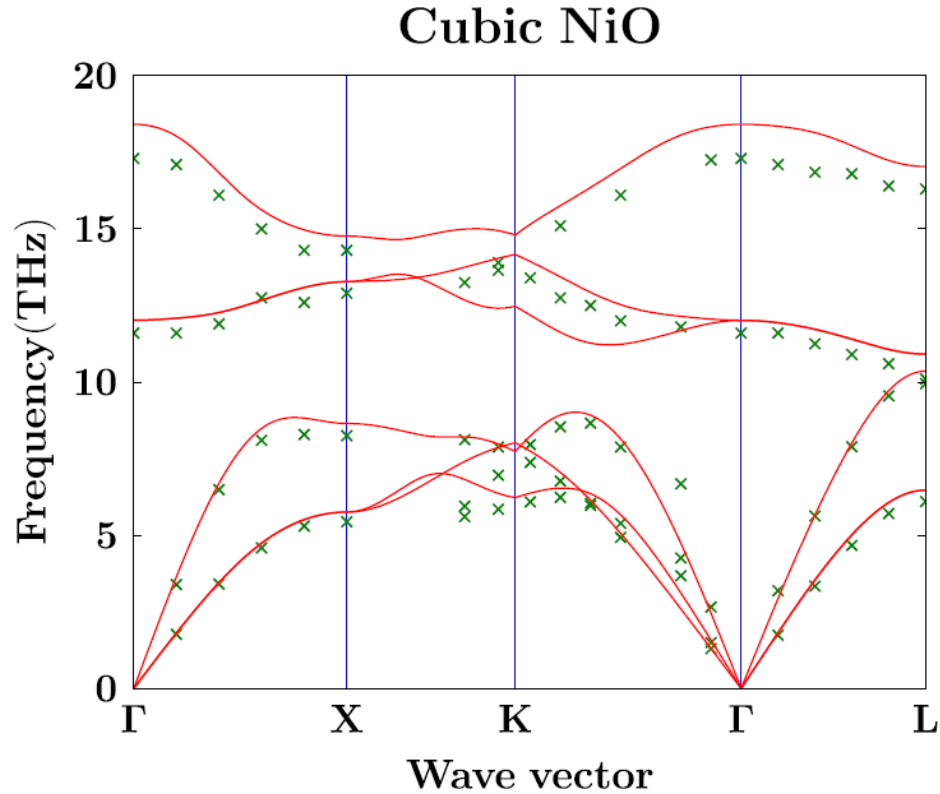
Magnetic primitive cell: $2 \times \text{Ni} + 2 \times \text{O}$ ($R-3m$)

DFT calculations only make sense when the magnetic primitive cell is used

Phonon dispersions and thermal conductivity can be studied in different two ways:

- (1) Using the magnetic primitive cell in the lattice dynamics and BTE part**
- (2) Neglecting the magnetic structure in the lattice dynamics and BTE, treating the crystal as an effective alloy of spin-up and spin-down atoms (analogous to the *virtual crystal approximation*)**

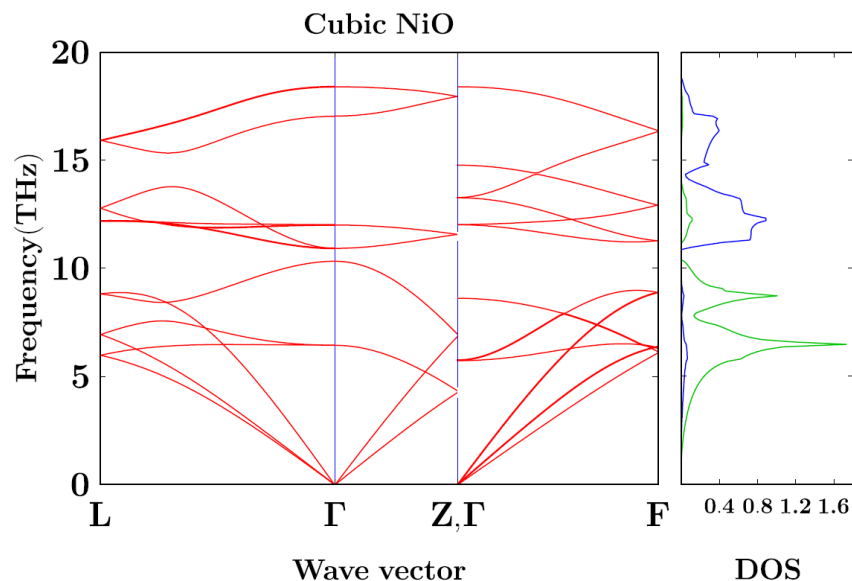
NiO phonon dispersions



¹ **Exp. data:** W. Reichardt, V. Wagner, and W. Kress, *J. Phys. C* **1975**, 8, 3955.

NiO Phonon dispersions for cubic and trigonal primitive cell

"Implicit magnetism" for Phonopy



"Explicit magnetism" for Phonopy

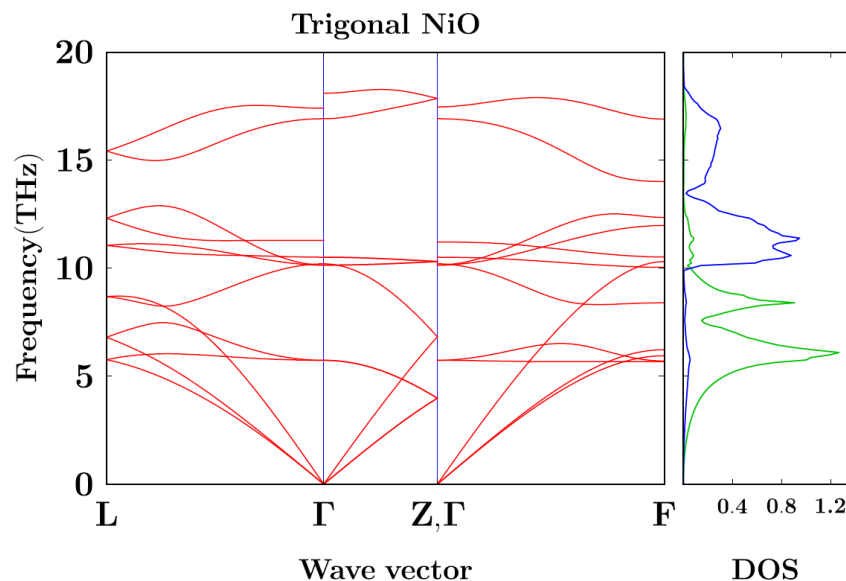
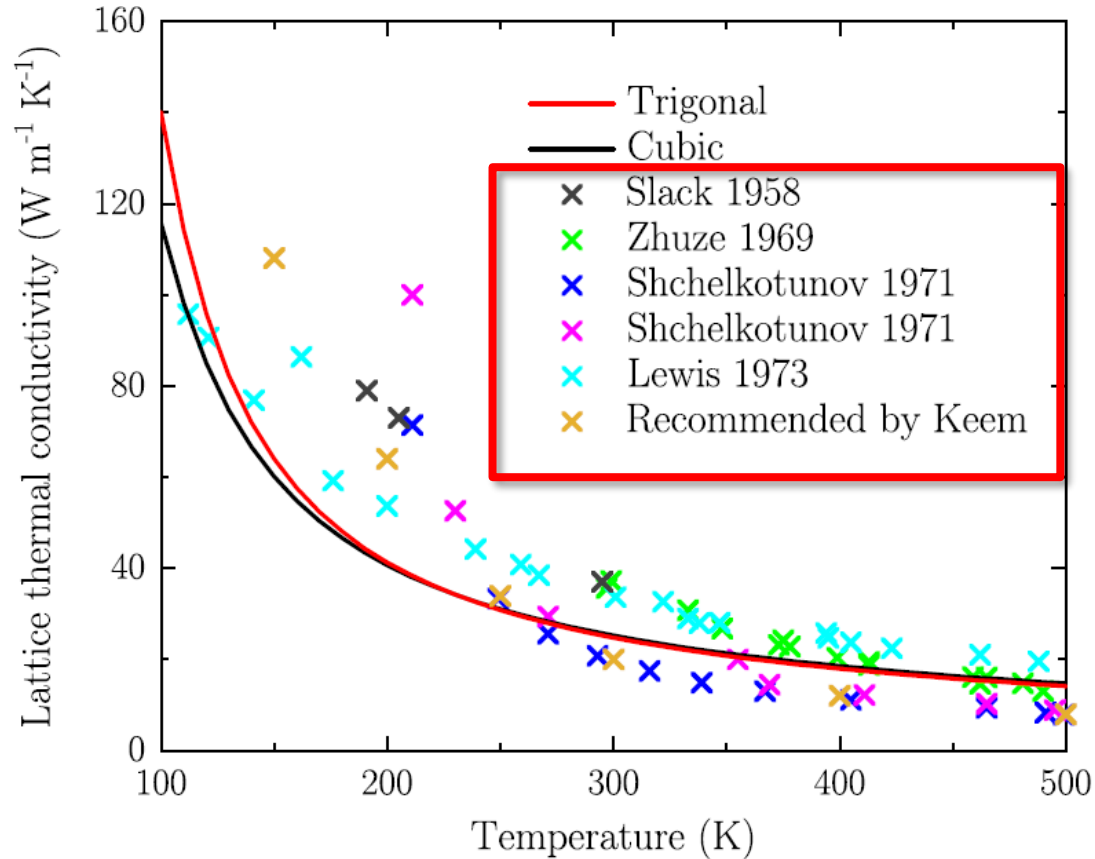


FIG. 3. Phonon dispersions and atom-projected phonon density of states (DOS) of NiO using two different primitive cells. The phonon dispersions of the cubic primitive cell were plotted in directions corresponding to the trigonal primitive cell, folded, and labeled according to the trigonal case. The following reciprocal space points were used for the cubic primitive cell: $L = (-1/2, 1/2, 1/2)$, $Z = (1/2, 1/2, 1/2)$, and $F = (-1/2, 1/2, 0)$. Corresponding points for the trigonal primitive cell are: $L = (0, 1/2, 0)$, $Z = (1/2, 1/2, 1/2)$, and $F = (1/2, 1/2, 0)$. In the DOS plots the green and blue lines correspond to Ni and O contributions, respectively.

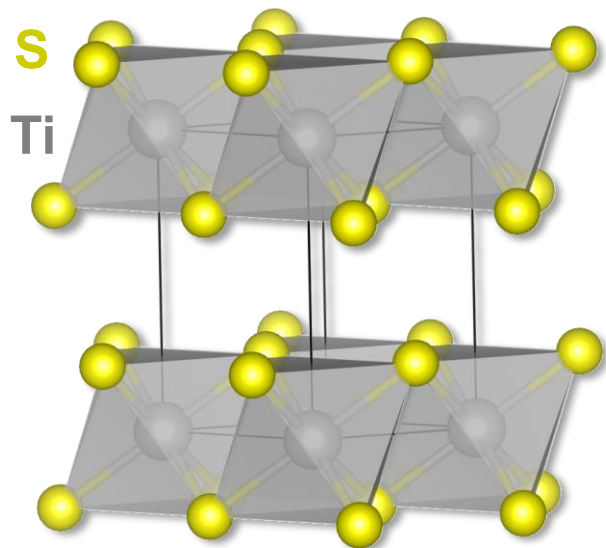
NiO lattice thermal conductivity



**Experimental data
vary a lot**

Appendix 2: TiS_2 , ZrS_2 , and HfS_2

Computational details for TiS_2 , ZrS_2 , HfS_2



TiS_2 ($P\text{-}3m$)
3 atoms in the unit cell

PBE0/TZVP

Empirical DFT-D3 dispersion correction

Zero-Damping (ZD) and Becke-Johnson (BJ) damping tested for D3

Damping function affects the behavior of the dispersion correction

Harmonic force constants: 5x5x3 (Ti) or 3x3x2 (Zr, Hf)

Anharmonic FCs: 3x3x2 supercell, no distance cutoffs

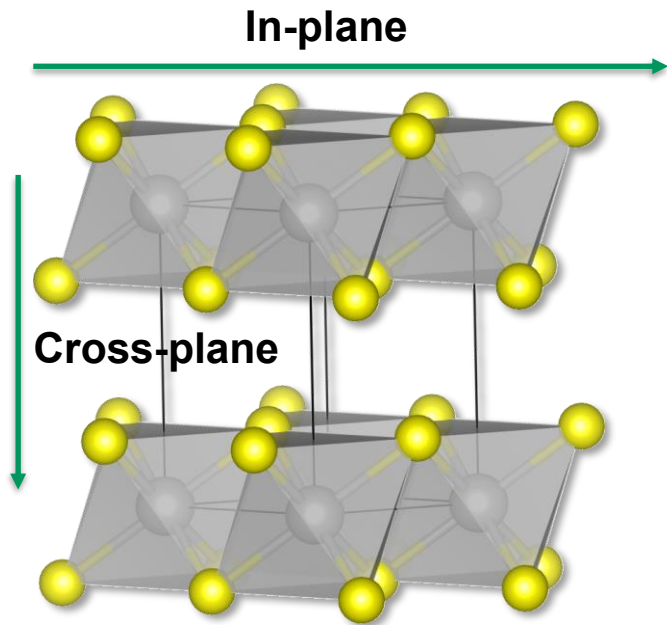
The role of the D3 dispersion correction

Table 1. DFT-Optimized and Experimental^{52,53} Lattice Constants a and c (in Å) for TiS₂, ZrS₂, and HfS₂^a

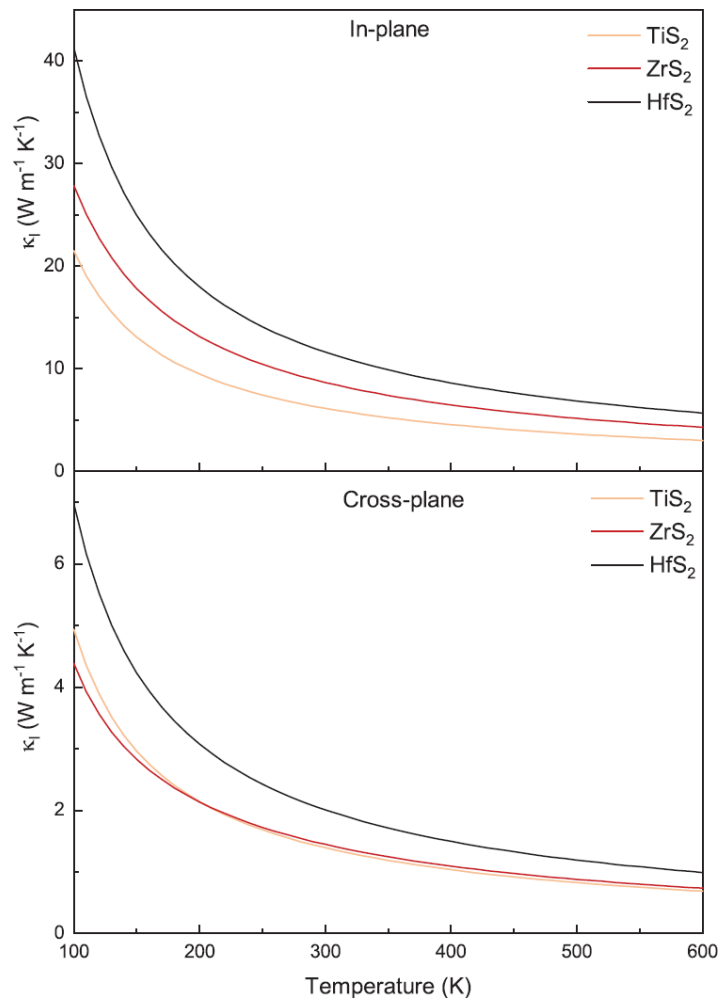
param.	TiS ₂	ZrS ₂	HfS ₂
PBE0			
a	3.41 (+0.0%)	3.68 (+0.5%)	3.68 (+1.3%)
c	6.22 (+9.0%)	6.43 (+10.5%)	6.67 (+14.2%)
PBE0-D3(BJ)			
a	3.36 (−1.4%)	3.63 (−0.8%)	3.64 (+0.1%)
c	5.51 (−3.3%)	5.62 (−3.4%)	5.65 (−3.1%)
PBE0-D3(ZD)			
a	3.40 (−0.4%)	3.66 (−0.1%)	3.66 (+0.6%)
c	5.67 (−0.7%)	5.73 (−1.5%)	5.73 (−1.8%)

Deviation from experimental crystal structure shown in parentheses

Lattice thermal conductivity of MS_2



Unexpected result: HfS_2 with the largest mass has the highest κ_l

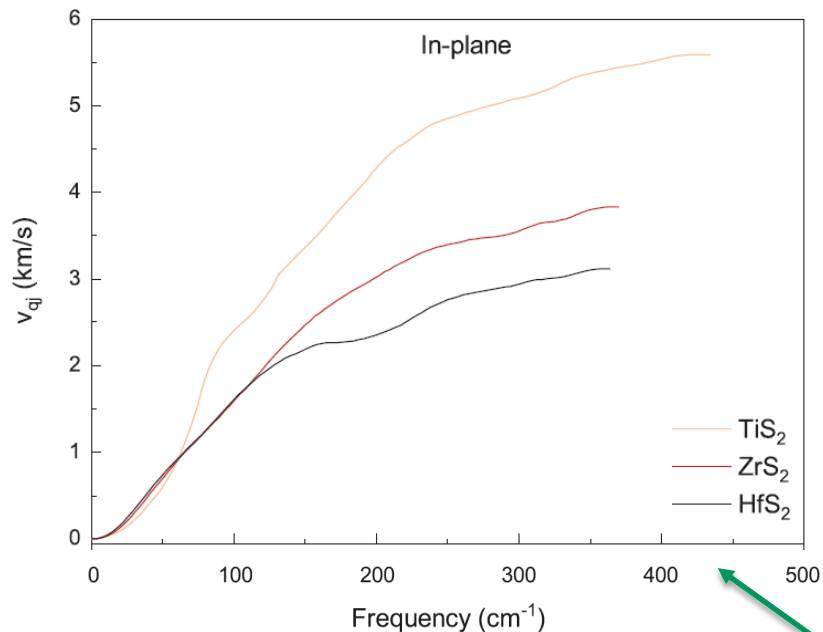


TiS_2 κ_l at 300 K
($\text{W m}^{-1} \text{K}^{-1}$)
Exp.: 4.5–6.4
DFT: 6.1

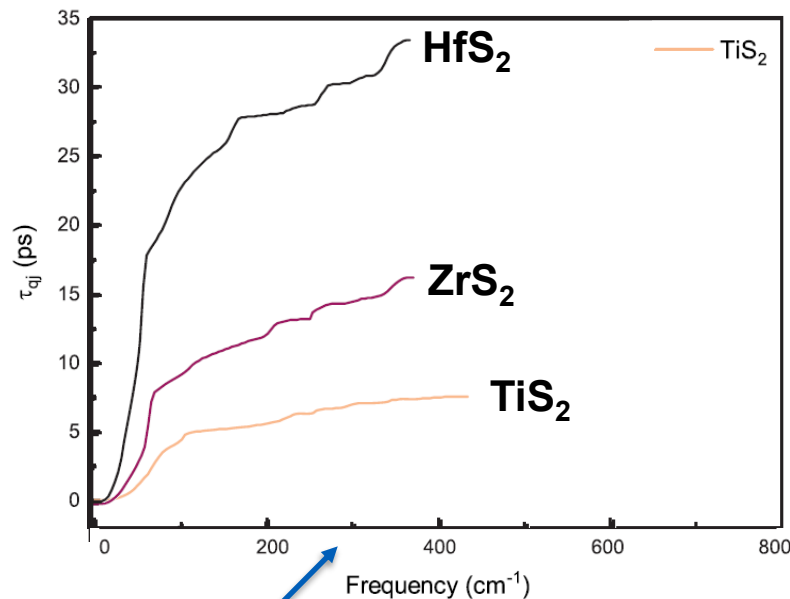
TiS_2 κ_t at 300 K
($\text{W m}^{-1} \text{K}^{-1}$)
Exp.: 4.2
DFT: 1.4

HfS₂ has the longest phonon lifetimes (“least anharmonic”)

Cumulative phonon group velocity



Cumulative phonon lifetime



$$\kappa = \frac{1}{NV_0} \sum_{\lambda} C_{\lambda} \mathbf{v}_{\lambda} \otimes \mathbf{v}_{\lambda} \tau_{\lambda}^{\text{SMRT}}$$